



Report 3

Secure Internal Office Network Using VLAN

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**NETWORK SECURITY –
BATCH 2**



Introduction

The objective of this task was to secure an internal office network by implementing Virtual Local Area Networks (VLANs). The network was divided into separate departments (HR and IT) to improve security and reduce unnecessary communication between users. A Cisco router, Layer 2 switch, and Wireshark were used to configure, monitor, and verify the network.

Objectives

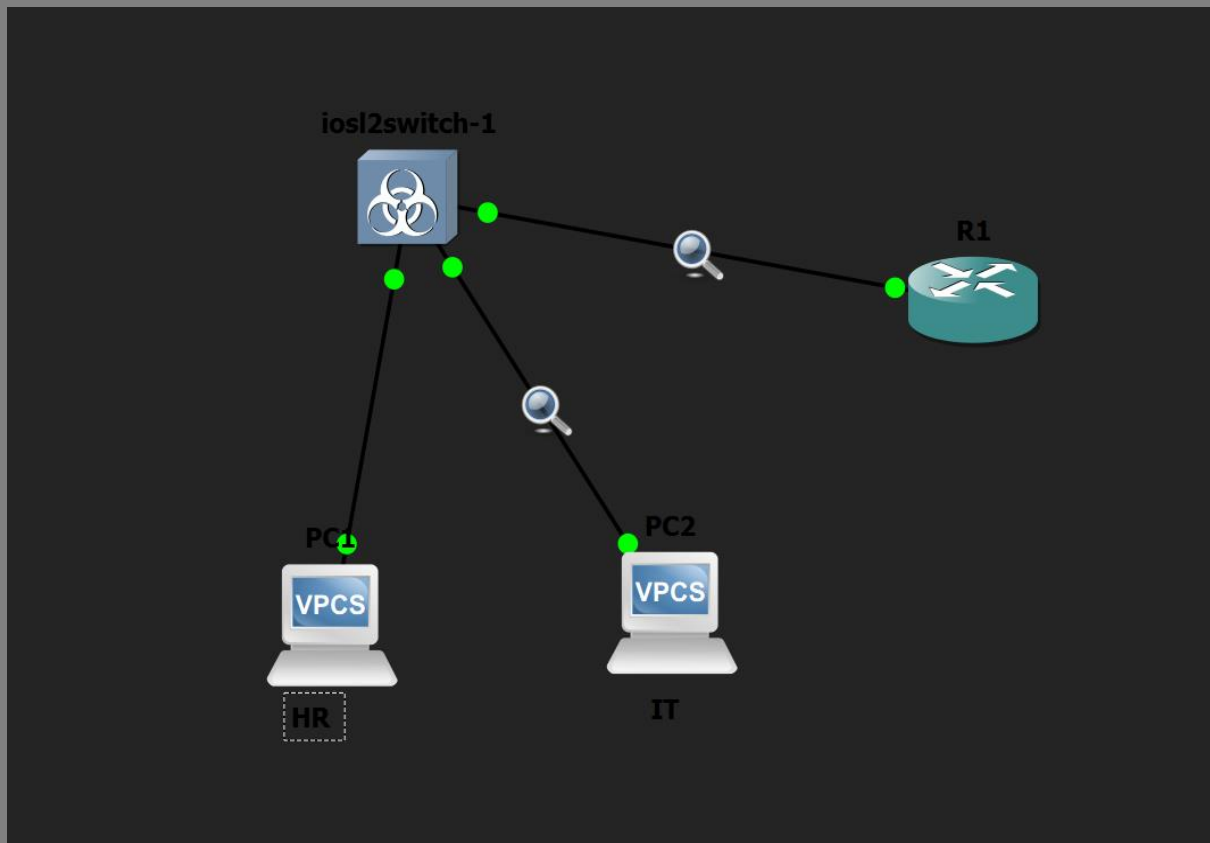
- Create separate VLANs for HR, IT, departments.
- Configure access ports on the Layer 2 switch.
- Configure Router-on-a-Stick for inter-VLAN routing.
- Test communication between VLANs.
- Apply communication restrictions using an Access Control List (ACL).
- Capture and analyze network traffic using Wireshark.

Software and Tools Used:

- GNS3
- Cisco c7200 Router
- Cisco IOU Layer 2 Switch
- VPCS
- Wireshark

Network Topology:

The network consists of one Cisco c7200 router connected to one Layer 2 switch through a trunk link. Two PCs representing HR, IT departments are connected to separate switch ports.



VLAN Configuration:

Two VLANs were created on the switch.

VLAN ID	VLAN Name	Port
10	HR	Ethernet0/1
20	IT	Ethernet0/2

The link between the router and switch was configured as an IEEE 802.1Q trunk. (Screenshot before Assigning IP-Addr)

```

*****
vIOS-L2-01>enable
vIOS-L2-01#show ip int brief
Interface      IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0  unassigned      YES unset  down           down
GigabitEthernet0/1  unassigned      YES unset  down           down
GigabitEthernet0/2  unassigned      YES unset  down           down
vIOS-L2-01#vlan 10
^

```

Creation of three VLANs on the Layer 2 switch:

```
^
vIOS-L2-01#config t
Enter configuration commands, one per line. End with CNTL/Z.
vIOS-L2-01(config)#vlan 10
vIOS-L2-01(config-vlan)#name HR
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 20
vIOS-L2-01(config-vlan)#name IT
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#vlan 30
vIOS-L2-01(config-vlan)#name Admin
vIOS-L2-01(config-vlan)#exit
vIOS-L2-01(config)#int e0/1
      ^
```

Configuration of switch access ports:

```
vIOS-L2-01(config)#int g0/1
vIOS-L2-01(config-if)#switchport mode access
vIOS-L2-01(config-if)#switchport access vlan 10
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#
vIOS-L2-01(config)#
vIOS-L2-01(config)#int g0/2
vIOS-L2-01(config-if)#switchport mode acces
vIOS-L2-01(config-if)#switchport access vlan 20
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#
```

Trunk Port Configuration:

```
vIOS-L2-01(config)#int g0/0
vIOS-L2-01(config-if)#switchport
*Jul  8 12:22:31.874: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
vIOS-L2-01(config-if)#switchport trunk encapsulation dot1q
vIOS-L2-01(config-if)#switchport mode trunk
vIOS-L2-01(config-if)#no shutdown
vIOS-L2-01(config-if)#exit
vIOS-L2-01(config)#end
vIOS-L2-01#wr
```

Show Vlan Brief:

```
vIOS-L2-01#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	HR	active	Gi0/1
20	IT	active	Gi0/2
30	Admin	active	
100	VLAN100	active	
200	VLAN0200	active	
300	VLAN0300	active	
1002	fddi-default	act/unsup	
1003	trcrf-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trbrf-default	act/unsup	

```
vIOS-L2-01#
```

Access-list:

```
vIOS-L2-01#show access-list
```

```
Extended IP access list preauth_ipv4_acl (per-user)
```

```
10 permit udp any any eq domain
20 permit tcp any any eq domain
30 permit udp any any eq bootps any
40 permit udp any any eq bootpc
50 permit udp any any eq bootpc any
60 deny ip any any
```

```
IPv6 access list preauth_ipv6_acl (per-user)
```

```
permit udp any any eq domain sequence 10
permit tcp any any eq domain sequence 20
permit icmp any any nd-ns sequence 30
permit icmp any any nd-na sequence 40
permit icmp any any router-solicitation sequence 50
permit icmp any any router-advertisement sequence 60
permit icmp any any redirect sequence 70
permit udp any eq 547 any eq 546 sequence 80
permit udp any eq 546 any eq 547 sequence 90
deny ipv6 any any sequence 100
```

```
vIOS-L2-01#
```

Router Configuration: (Before Assigning Interface)

```
R1#show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	manual	up	up
Serial2/0	unassigned	YES	unset	administratively down	down
Serial2/1	unassigned	YES	unset	administratively down	down
Serial2/2	unassigned	YES	unset	administratively down	down
Serial2/3	unassigned	YES	unset	administratively down	down

```
R1#
```

Router-on-a-Stick was configured using FastEthernet0/0 with two sub interfaces. Each VLAN was assigned its own gateway for communication.

Sub interface	VLAN	IP Address
Fa0/0.10	VLAN 10	192.168.10.1
Fa0/0.20	VLAN 20	192.168.20.1

Router-on-a-Stick Configuration (VLAN 10):

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#int f0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip add 192.168.10.1 255.255.255.0
R1(config-subif)#exit
R1(config)#
R1(config)#
```

Router-on-a-Stick Configuration (VLAN 20):

```
R1(config)#
R1(config)#
R1(config)#int f0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip add 192.168.20.1 255.255.255.0
R1(config-subif)#exit
R1(config)#
R1(config)#end
R1#
*Jul  8 12:31:07.835: %SYS-5-CONFIG_I: Configured from console by console
R1#wr
Building configuration...
[OK]
R1#
```

Access Control List (ACL) Configuration:

```
R1#  
R1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
R1(config)# 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255  
R1(config)# 100 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255  
R1(config)#access-list 100 permit ip any any  
R1(config)#  
R1(config)#
```

Applying the ACL to Router Interfaces:

```
R1(config)#  
R1(config)#  
R1(config)#  
R1(config)#int f0/0.10  
R1(config-subif)#ip access-group 100 in  
R1(config-subif)#exit  
R1(config)#int f0/0.20  
R1(config-subif)#ip access-group 100 in  
R1(config-subif)#exit  
R1(config)#
```

Access List Verification:

```
vIOS-L2-01#show access-list  
Extended IP access list preauth_ipv4_acl (per-user)  
 10 permit udp any any eq domain  
 20 permit tcp any any eq domain  
 30 permit udp any eq bootps any  
 40 permit udp any any eq bootpc  
 50 permit udp any eq bootpc any  
 60 deny ip any any  
IPv6 access list preauth_ipv6_acl (per-user)  
 permit udp any any eq domain sequence 10  
 permit tcp any any eq domain sequence 20  
 permit icmp any any nd-ns sequence 30  
 permit icmp any any nd-na sequence 40  
 permit icmp any any router-solicitation sequence 50  
 permit icmp any any router-advertisement sequence 60  
 permit icmp any any redirect sequence 70  
 permit udp any eq 547 any eq 546 sequence 80  
 permit udp any eq 546 any eq 547 sequence 90  
 deny ipv6 any any sequence 100  
vIOS-L2-01#
```

PC Configuration:

Each department was assigned a different IP address.

Department	IP Address	Gateway
HR	192.168.10.10	192.168.10.1
IT	192.168.20.10	192.168.20.1

Communication Testing:

Connectivity tests were performed using the ping command.

Results:

```
PC1> ip 192.168.10.10/24 192.168.10.1
Checking for duplicate address...
PC1 : 192.168.10.10 255.255.255.0 gateway 192.168.10.1

PC1> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=134.036 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=80.162 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=106.577 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=241.896 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=150.287 ms
```

HR to IT – Blocked using ACL:

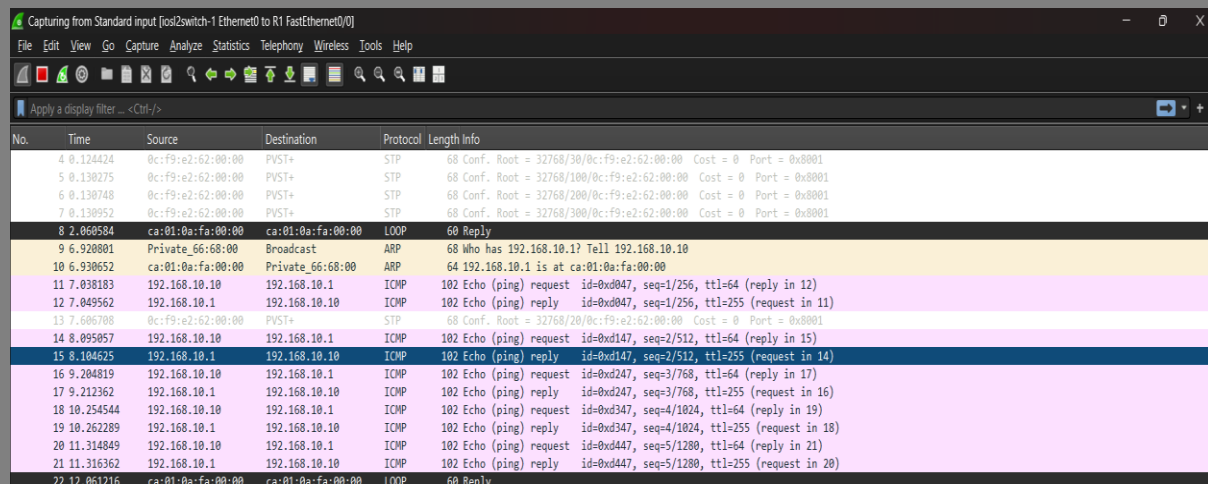
```
PC1> ping 192.168.20.10
*192.168.10.1 icmp_seq=1 ttl=255 time=155.828 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=2 ttl=255 time=154.895 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=3 ttl=255 time=66.276 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=4 ttl=255 time=101.727 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.10.1 icmp_seq=5 ttl=255 time=43.753 ms (ICMP type:3, code:13, Communication administratively prohibited)

PC1>
```

Wireshark Analysis:

Wireshark was used to capture ICMP traffic generated during the ping tests. The packet capture showed ICMP Echo Requests and ICMP Destination Unreachable messages when communication between restricted VLANs was attempted. This confirmed that the ACL was successfully enforcing communication restrictions.

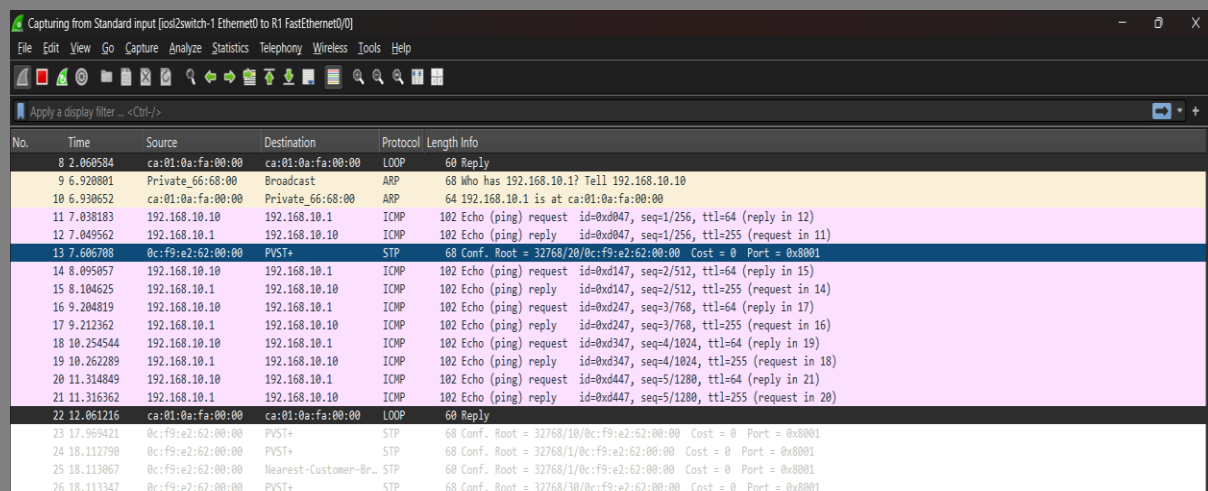
Packet Capture:



No.	Time	Source	Destination	Protocol	Length	Info
4	0.124424	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/30/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
5	0.130275	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/100/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
6	0.130748	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/200/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
7	0.130952	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/300/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
8	2.060584	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
9	6.920801	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.10.1? Tell 192.168.10.10
10	6.930652	ca:01:0a:fa:00:00	Private_66:68:00	ARP	64	192.168.10.1 is at ca:01:0a:fa:00:00
11	7.038183	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd047, seq=1/256, ttl=64 (reply in 12)
12	7.049562	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd047, seq=1/256, ttl=255 (request in 11)
13	7.606708	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
14	8.095057	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd147, seq=2/512, ttl=64 (reply in 15)
15	8.104625	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd147, seq=2/512, ttl=255 (request in 14)
16	9.204819	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd247, seq=3/768, ttl=64 (reply in 17)
17	9.212362	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd247, seq=3/768, ttl=255 (request in 16)
18	10.254544	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd347, seq=4/1024, ttl=64 (reply in 19)
19	10.262289	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd347, seq=4/1024, ttl=255 (request in 18)
20	11.314849	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd447, seq=5/1280, ttl=64 (reply in 21)
21	11.316362	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd447, seq=5/1280, ttl=255 (request in 20)
22	12.061216	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply

Wireshark Capture – STP and ARP Traffic:

This Wireshark capture shows Spanning Tree Protocol (STP) and Address Resolution Protocol (ARP) packets. STP packets are exchanged between network devices to prevent switching loops and maintain a stable network topology.



No.	Time	Source	Destination	Protocol	Length	Info
8	2.060584	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
9	6.920801	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.10.1? Tell 192.168.10.10
10	6.930652	ca:01:0a:fa:00:00	Private_66:68:00	ARP	64	192.168.10.1 is at ca:01:0a:fa:00:00
11	7.038183	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd047, seq=1/256, ttl=64 (reply in 12)
12	7.049562	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd047, seq=1/256, ttl=255 (request in 11)
13	7.606708	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
14	8.095057	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd147, seq=2/512, ttl=64 (reply in 15)
15	8.104625	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd147, seq=2/512, ttl=255 (request in 14)
16	9.204819	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd247, seq=3/768, ttl=64 (reply in 17)
17	9.212362	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd247, seq=3/768, ttl=255 (request in 16)
18	10.254544	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd347, seq=4/1024, ttl=64 (reply in 19)
19	10.262289	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd347, seq=4/1024, ttl=255 (request in 18)
20	11.314849	192.168.10.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd447, seq=5/1280, ttl=64 (reply in 21)
21	11.316362	192.168.10.1	192.168.10.10	ICMP	102	Echo (ping) reply id=0xd447, seq=5/1280, ttl=255 (request in 20)
22	12.061216	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
23	17.969421	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/10/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
24	18.112790	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/1/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
25	18.113067	0c:f9:e2:62:00:00	Nearest-Customer-Bn.	STP	68	Conf. Root = 32768/1/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
26	18.113347	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/30/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001

Wireshark Capture – Successful ICMP Communication:

This screenshot shows a successful ICMP (Ping) communication between the host and its default gateway. The ICMP Echo Request packets are followed by ICMP Echo Reply packets, confirming that the device can successfully communicate with its gateway.

221	323.584919	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
222	332.831907	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
223	333.643192	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
224	343.814734	Private_66:68:01	Broadcast	ARP	68	Who has 192.168.20.1? Tell 192.168.20.10
225	343.821804	ca:01:0a:fa:00:00	Private_66:68:01	ARP	64	192.168.20.1 is at ca:01:0a:fa:00:00
226	343.868330	192.168.20.10	192.168.20.1	ICMP	102	Echo (ping) request id=0x2049, seq=1/256, ttl=64 (reply in 227)
227	343.874152	192.168.20.1	192.168.20.10	ICMP	102	Echo (ping) reply id=0x2049, seq=1/256, ttl=255 (request in 226)
228	343.648155	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
229	344.144361	192.168.20.10	192.168.20.1	ICMP	102	Echo (ping) request id=0x2149, seq=2/512, ttl=64 (reply in 230)
230	344.150145	192.168.20.1	192.168.20.10	ICMP	102	Echo (ping) reply id=0x2149, seq=2/512, ttl=255 (request in 229)
231	345.214227	192.168.20.10	192.168.20.1	ICMP	102	Echo (ping) request id=0x2249, seq=3/768, ttl=64 (reply in 232)
232	345.221838	192.168.20.1	192.168.20.10	ICMP	102	Echo (ping) reply id=0x2249, seq=3/768, ttl=255 (request in 231)
233	345.961106	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/10/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
234	345.963555	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/10/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001

Wireshark Capture – ACL Blocking Communication:

This Wireshark capture shows a failed communication attempt between different VLANs. ICMP Echo Request packets were sent, but instead of receiving Echo Replies, the router generated ICMP Destination Unreachable (Communication administratively filtered) messages.

180	264.766557	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/300/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001
181	269.066503	192.168.20.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd648, seq=1/256, ttl=64 (no response found!)
182	269.073614	192.168.20.1	192.168.20.10	ICMP	74	Destination unreachable (Communication administratively filtered)
183	270.097247	192.168.20.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd748, seq=2/512, ttl=64 (no response found!)
184	270.104446	192.168.20.1	192.168.20.10	ICMP	74	Destination unreachable (Communication administratively filtered)
185	271.225305	192.168.20.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd848, seq=3/768, ttl=64 (no response found!)
186	271.231240	192.168.20.1	192.168.20.10	ICMP	74	Destination unreachable (Communication administratively filtered)
187	272.296898	192.168.20.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xd948, seq=4/1024, ttl=64 (no response found!)
188	272.303046	192.168.20.1	192.168.20.10	ICMP	74	Destination unreachable (Communication administratively filtered)
189	272.612984	ca:01:0a:fa:00:00	ca:01:0a:fa:00:00	LOOP	60	Reply
190	273.464568	192.168.20.10	192.168.10.1	ICMP	102	Echo (ping) request id=0xda48, seq=5/1280, ttl=64 (no response found!)
191	273.474529	192.168.20.1	192.168.20.10	ICMP	74	Destination unreachable (Communication administratively filtered)
192	273.870409	0c:f9:e2:62:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:f9:e2:62:00:00 Cost = 0 Port = 0x8001

ARP Table Verification on PC2:

This screenshot shows the ARP table on PC2 before and after communication with the default gateway. Initially, the ARP table was empty because no communication had occurred. After a successful ping to the gateway (192.168.20.1), the ARP table was updated with the gateway's MAC address.

```
PC2> show arp
arp table is empty

PC2> ping 192.168.10.1
*192.168.20.1 icmp_seq=1 ttl=255 time=87.618 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.20.1 icmp_seq=2 ttl=255 time=92.813 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.20.1 icmp_seq=3 ttl=255 time=14.906 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.20.1 icmp_seq=4 ttl=255 time=106.918 ms (ICMP type:3, code:13, Communication administratively prohibited)
*192.168.20.1 icmp_seq=5 ttl=255 time=81.231 ms (ICMP type:3, code:13, Communication administratively prohibited)

PC2> ping 192.168.20.1
84 bytes from 192.168.20.1 icmp_seq=1 ttl=255 time=93.044 ms
84 bytes from 192.168.20.1 icmp_seq=2 ttl=255 time=79.047 ms
84 bytes from 192.168.20.1 icmp_seq=3 ttl=255 time=28.061 ms
84 bytes from 192.168.20.1 icmp_seq=4 ttl=255 time=338.805 ms
84 bytes from 192.168.20.1 icmp_seq=5 ttl=255 time=113.794 ms

PC2> show arp
ca:01:0a:fa:00:00 192.168.20.1 expires in 92 seconds

PC2> █
```

ARP Table Verification on Router:

This screenshot shows the router's ARP table using the show arp command. The router has successfully learned the MAC addresses of devices connected to VLAN 10 and VLAN 20 through ARP. The ARP entries confirm that the router can correctly map IP addresses to MAC addresses, allowing communication within each VLAN while enforcing the configured ACL policies between restricted VLANs.

```
R1#show arp
Protocol Address      Age (min) Hardware Addr  Type  Interface
Internet 192.168.10.1      -    ca01.0afa.0000  ARPA  FastEthernet0/0.10
Internet 192.168.10.10    2    0050.7966.6800  ARPA  FastEthernet0/0.10
Internet 192.168.20.1      -    ca01.0afa.0000  ARPA  FastEthernet0/0.20
R1# █
```

Security Benefits:

- Improved network security through VLAN segmentation.
- Reduced broadcast traffic.
- Protected departmental resources from unauthorized access.
- Easier network management and troubleshooting.
- Better control of communication between departments using ACLs.

Challenges Faced:

During the configuration process, I initially faced difficulties understanding VLAN configuration, trunk links, and ACL implementation. I also encountered interface configuration issues and communication problems between VLANs. These challenges were resolved through troubleshooting, verification commands, and guidance from my internship supervisor.

Learning Outcomes:

Through this task, I gained practical experience in configuring VLANs, assigning switch ports, implementing Router-on-a-Stick, configuring Access Control Lists (ACLs), and analysing network traffic using Wireshark. I also learned how VLAN segmentation enhances network security and improves communication control within an organization.

Conclusion:

This task successfully demonstrated how VLANs can be used to secure an internal office network. By separating HR, IT, and Admin into different VLANs and applying ACLs, unauthorized communication between departments was restricted while maintaining proper network functionality. Wireshark packet analysis further verified the effectiveness of the implemented security policies. The task enhanced my practical knowledge of enterprise networking and network security concepts.